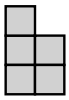
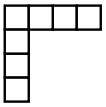
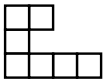
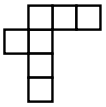
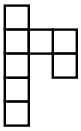
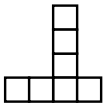
Preview: y2007_q08 (v1)

Completing a rectangle: the missing part has a fixed shape and size; turn pieces in your head

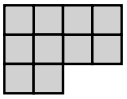
Part A: Each question is worth 3 points

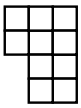
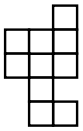
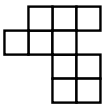
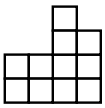
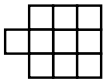
1. Which of the five pieces fits together with the grey piece to make a rectangle? Pieces may be turned and flipped over.



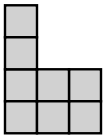
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

2. Which of the five pieces fits together with the grey piece to make a rectangle? Pieces may be turned and flipped over.



A.  ***B.***  ***C.***  ***D.***  ***E.*** 

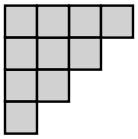
3. Tom wants to add one more piece to the grey piece to make the smallest possible rectangle.



How many little squares must that piece have?

A. 2 ***B.*** 3 ***C.*** 4 ***D.*** 8 ***E.*** 12

4. Tom wants to add one more piece to the grey piece to make the smallest possible rectangle.

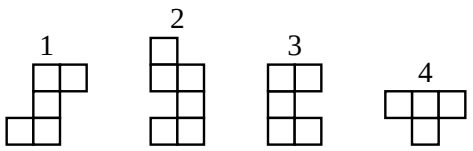


How many little squares must that piece have?

A. 6 ***B.*** 5 ***C.*** 4 ***D.*** 3 ***E.*** 2

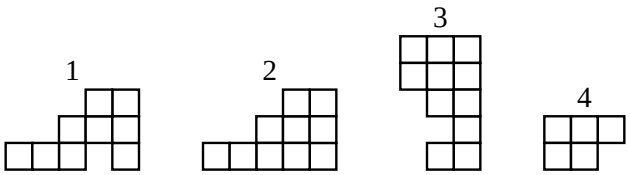
Part B: Each question is worth 4 points

5. Which two of these pieces fit together to make a rectangle? Pieces may be turned and flipped over.



A. 2 and 3 ***B.*** 1 and 4 ***C.*** 1 and 3 ***D.*** 3 and 4 ***E.*** 1 and 2

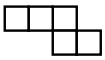
6. Which two of these pieces fit together to make a rectangle? Pieces may be turned and flipped over.



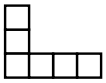
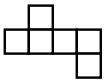
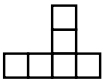
A. 2 and 4 ***B.*** 1 and 4 ***C.*** 3 and 4 ***D.*** 1 and 2 ***E.*** 2 and 3

Part C: Each question is worth 5 points

7. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

A.  ***B.*** ***C.*** ***D.*** ***E.***

8. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

A. ***B.***  ***C.*** ***D.***  ***E.*** 

Answer Key

1. **B**	**2. A**	**3. C**	**4. A**	**5. D**	**6. A**
7. **C**	**8. A**				

Solutions

1. B — The grey piece and the right piece together fill a 3 × 4 rectangle: 12 – 5 = 7 squares, in exactly the shape of the gap. Piece B (turned) fills it; the others have a square too many, too few, or in the wrong place. [2007 Q8 · complete_rectangle]

2. A — The grey piece and the right piece together fill a 4 × 5 rectangle: 20 – 10 = 10 squares, in exactly the shape of the gap. Piece A (turned) fills it; the others have a square too many, too few, or in the wrong place. [2007 Q8 · complete_rectangle]

3. C (4) — The smallest rectangle around the grey piece is 3 × 4 = 12 squares. 12 – 8 = 4. [2007 Q8 · squares_missing]

4. A (6) — The smallest rectangle around the grey piece is 4 × 4 = 16 squares. 16 – 10 = 6. [2007 Q8 · squares_missing]

5. D (3 and 4) — Pieces 3 and 4 have 9 squares together and fit into a 3 × 3 rectangle. No other pair does. [2007 Q8 · two_make_rectangle]

6. A (2 and 4) — Pieces 2 and 4 have 15 squares together and fit into a 5 × 3 rectangle. No other pair does. [2007 Q8 · two_make_rectangle]

7. C — Two copies have 10 squares, so the rectangle is 1 × 10 or 2 × 5. Only two copies of piece C fit together like that. [2007 Q8 · two_copies]

8. A — Two copies have 12 squares, so the rectangle is 1 × 12 or 2 × 6 or 3 × 4. Only two copies of piece A fit together like that. [2007 Q8 · two_copies]